

Madhu Chandrika Balram

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EDUCATION

Master's in Computer Science, DePaul University 3.71 GPA	Sep 2024 – Jun 2026 Chicago, USA
Bachelor of Technology in Electronics and Communication, REVA University 3.7 GPA	Aug 2018 – Jul 2022 Bangalore, India

PROFESSIONAL EXPERIENCE

AI Agent Developer Intern, NeuralSeek <i>Technologies: AI Agents · LLMs · RAG · Application Workflow Integration · Prompt Engineering</i> - Built enterprise-focused AI agents integrated into task-oriented software application workflows using large language models. - Implemented retrieval-augmented generation (RAG) by connecting agents to structured knowledge bases, improving response accuracy and grounding. - Designed agent workflows and prompt logic to generate structured, application-ready outputs consumable by software systems. - Applied reliability and governance patterns, including structured response schemas and validation checks, to support controlled, enterprise-ready AI usage.	Oct 2025 – Nov 2025 Chicago, Illinois
Custom Software Engineering Associate, Accenture <i>Technologies: Java · REST APIs · SQL · Relational Databases · Backend Development · Cloud Applications (Heroku)</i> - Developed and enhanced backend components for enterprise Java-based applications supporting core business workflows. - Implemented and modified RESTful APIs, application logic, and data access layers as part of feature development and system enhancements. - Worked with relational databases (SQL) to support data-driven backend functionality and application workflows. - Supported cloud-deployed backend applications by analyzing logs and runtime behavior to diagnose and resolve functional issues in development and production environments.	Dec 2022 – Jul 2024 Bangalore, India
Programmer Analyst Trainee Intern, Cognizant <i>Technologies: Java · SQL · Relational Databases · Selenium · Backend Development · Test Automation</i> - Developed Java-based backend application logic as part of enterprise software systems, contributing to feature implementation. - Worked with relational databases (SQL) to support backend workflows and validate business logic across application layers. - Implemented test automation using Java and Selenium to support regression testing and improve code reliability.	Feb 2022 – Jul 2022 Bangalore, India

PROJECTS

Aspire-AI — AI-Powered Career Assistant, Technologies: Next.js · React · TypeScript · REST APIs · Generative AI Integration	Jun 2025 – Aug 2025
- Built a full-stack web application integrating generative AI to support resume creation, interview preparation, and career guidance workflows. - Implemented server-side logic and REST API routes using Next.js to generate structured, role-specific content through large language model integration.	
Real Estate Management System, Technologies: Java · Spring Boot · Spring Security · Spring Data JPA · MySQL · REST APIs	Apr 2025 – Jun 2025
- Built backend services using Java and Spring Boot, exposing REST APIs to manage property listings and user workflows. - Implemented role-based access control using Spring Security and designed persistence layers with Spring Data JPA and MySQL.	
E-Commerce Product Store, Technologies: Node.js · Express.js · React · MongoDB · REST APIs · Stripe	Mar 2025 – Apr 2025
- Developed a full-stack e-commerce application with RESTful backend services to support product browsing, user management, and order workflows. - Implemented authentication, authorization, and secure payment processing to enable end-to-end checkout functionality.	

TECHNICAL SKILLS

Programming Language: Java · JavaScript · TypeScript · Python · SQL
Backend & Framework: Spring Boot · Spring Security · Spring Data JPA · Node.js · Express.js
Frontend Technologies: React.js · Next.js · HTML · CSS / Tailwind CSS
APIs & Databases: REST APIs · MySQL · MongoDB
Security: Authentication & Authorization
Applied AI: Generative AI Integration

CERTIFICATE

Master Spring Boot 3 & Spring Framework 6 with Java — Udemy
The Complete Full-Stack Web development Bootcamp- Udemy